



Explicitly integrating MeSH thesaurus help into health information retrieval systems: An empirical user study

Xiangming Mu ^{a,*}, Kun Lu ^b, Hohyon Ryu ^c

^a School of Information Studies, University of Wisconsin–Milwaukee, Milwaukee, USA

^b School of Information Management, Wuhan University, Wuhan, China

^c School of Information, University of Texas at Austin, Austin, USA

ARTICLE INFO

Article history:

Received 31 October 2009

Received in revised form 28 May 2012

Accepted 6 March 2013

Available online 31 July 2013

Keywords:

Health information retrieval

MeSH terms

MeSH Browser

Usability

User experience

Interface design

ABSTRACT

When consumers search for health information, a major obstacle is their unfamiliarity with the medical terminology. Even though medical thesauri such as the Medical Subject Headings (MeSH) and related tools (e.g., the MeSH Browser) were created to help consumers find medical term definitions, the lack of direct and explicit integration of these help tools into a health retrieval system prevented them from effectively achieving their objectives. To explore this issue, we conducted an empirical study with two systems: One is a simple interface system supporting query-based searching; the other is an augmented system with two new components supporting MeSH term searching and MeSH tree browsing. A total of 45 subjects were recruited to participate in the study. The results indicated that the augmented system is more effective than the simple system in terms of improving user-perceived topic familiarity and question-answer performance, even though we did not find users spend more time on the augmented system. The two new MeSH help components played a critical role in participants' health information retrieval and were found to allow them to develop new search strategies. The findings of the study enhanced our understanding of consumers' search behaviors and shed light on the design of future health information retrieval systems.

Published by Elsevier Ltd.

1. Introduction

Using the Internet to look for health information is a common behavior for consumers. According to the Pew Internet and American Life Project report, over 80% of American adults have searched for health information online (Fox, 2006). Yet, due to complex and sophisticated terminology, health information seekers, especially laypersons, often have difficulty in finding the health-related information they seek (Cline & Haynes, 2001; Hersh, 2009; Keselman, Browne, & Kaufman, 2008; Soergel, Tse, & Slaughter, 2004; Zeng, Kogan, Ash, Greenes, & Boxwala, 2002; Zhang, Wolfram, Wang, Hong, & Gillis, 2008). There are significant mismatches between the consumer terminology, and both the information source terminology and standard medical vocabularies (Zeng et al., 2002). Many health information retrieval systems rely on users to correctly phrase their queries in order to successfully retrieve related information (Yeganova, Comeau, Kim, & Wilbur, 2009). One main challenge, however, is that consumers may lack the domain knowledge needed to choose the right query phrases and their vocabulary may not match the vocabulary of the health area in which they are searching for information (Blair & Maron, 1985; Poikonen

* Corresponding author. Address: School of Information Studies, University of Wisconsin–Milwaukee, Northwest Quad Building B, 2025 E. Newport, Milwaukee, WI 53211, USA. Tel.: +1 (414) 229 4707; fax: +1 (414) 229 6699.

E-mail address: mux@uwm.edu (X. Mu).

& Vakkari, 2009; Voorhees, 1994; Zeng et al., 2006). To address this issue, a medical thesaurus is frequently used to solve the vocabulary mismatch problem (Gault, Schultz, & Davies 2002; Hersh, 2009; Shultz, 2006).

By providing controlled vocabularies, synonyms and closely related concepts, a medical thesaurus enables users to formulate queries that retrieve potentially relevant information not expressed in layperson terms. There are more than 150 medical thesauri in the Unified Medical Language System (UMLS) Metathesaurus. Among them, Medical Subject Headings (MeSH) is one of the most commonly used to facilitate health information retrieval (Yeganova et al., 2009).

MeSH is a tree-structured controlled vocabulary produced by the National Library of Medicine (NLM) to facilitate biomedical and health-related information retrieval. It bridges commonly used layperson terms with controlled medical terms used in professional literature. Despite its great potentials, MeSH thesaurus has not been explicitly integrated into a health information retrieval system. After finding a few good citations, a domain expert may successfully look at the MeSH thesaurus and formulate more relevant future queries with appropriate MeSH terms (Marchionini, Dwiggins, Katz, & Lin, 1993). The general users, particularly those who only search occasionally, however, may find it difficult to use current information retrieval systems to find appropriate MeSH terms without direct and explicit help from the MeSH thesaurus. In practice, one solution is to provide automatic MeSH term expansions, such as is implemented in PubMed.

PubMed is a text-based search and retrieval system developed by the National Center for Biotechnology Information (NCBI). According to the National Library of Medicine, in 2002 one third of PubMed's users came from the general public (Herskovic, Tantaka, Hersh, & Bernstam, 2007). To help consumers unfamiliar with the MeSH terms, a MeSH term expansion technique called "Automatic Term Mapping" was offered to automatically add related MeSH terms behind the scenes (Yeganova et al., 2009). For example, a user's plain text query "tumor in breast" will be modified as

"tumour"[All Fields] OR "neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "tumor"[All Fields]) AND ("breast"[MeSH Terms] OR "breast"[All Fields])

Regardless of its success, this technique has a key drawback. It does not provide users any definitions of the MeSH terms that have been expanded. If it did, users could have the opportunity to better understand and learn the preferred medical vocabularies used, and why documents containing the term "breast neoplasms" may appear on top of the result list for a search query of "breast cancer".

Another commonly used MeSH thesaurus help component is the MeSH Browser which is an online MeSH vocabulary look-up aid provided by NLM to help quickly locate MeSH descriptors of interest. A tree hierarchical structure related to the descriptors is available allowing users to identify narrower, broader, and closely related terms. While both PubMed and MeSH Browser could be very helpful to users, their usefulness may be hindered because each system exists independently of the other. When a user enters a query into PubMed, she/he is limited to using only that system and no references are made to information contained within the helpful MeSH Browser tool.

Integrating new MeSH help components, such as a MeSH Browser or a MeSH term definition tool into a health information retrieval system, however, may increase the complexity of the search interface. To explore the trade-off between the helpfulness of added components and a simple interface in facilitating health information searching, we conducted an empirical usability study. We used two health information retrieval systems. One is a simple interface with a traditional search system, which we call SimpleMed. The other is augmented with the MeSH Term Browser and MeSH Tree Browser help components, which we call MeshMed. We would like to explore how vocabulary-help components such as the MeSH Term Browser and Tree Browser impact consumer's behaviors and the effectiveness of health information retrieval. One motivation of the study is that we believe there are still a fraction of people who would like to use advanced search functions or other help components to achieve more accurate and relevant results when they search health information.

The subsequent sections of this paper are organized as follows: After a brief literature review, we first describe our research goals. Then we specify our study methodology, including descriptions of the two systems, the participants, the data corpus, search tasks, measurements, and study design and procedure. Our study findings include results of search effectiveness analysis of SimpleMed and MeshMed, new MeSH component contribution analysis, users' search behaviors and patterns, and participants' comment analysis. Finally, we conclude our paper with discussion, implications, and future work.

2. Literature review

Various thesauri have been studied in information retrieval (Aronson & Rindfleisch, 1997; Bhogal, Macfarlane, & Smith, 2007; Greenberg, 2001; Kristensen, 1993; Voorhees, 1994). Among them, the MeSH thesaurus is the most commonly used in the health domain (Srinivasan, 1996a, 1996b; Stokes, et al., 2009; Gobel, Andreatta, Masser, & Pfeiffer, 2001; Hersh, Price, & Donohoe, 2000; Lu, Kim, & Wilbur, 2009). To bridge the gap between consumer and professional medical vocabulary, currently there are three major research efforts regarding the use of the MeSH thesaurus: query expansion, the use of intermediary thesauri, and structure navigation.

Query expansion with the MeSH thesaurus usually involves the automatic addition of related MeSH terms to users' initial queries in order to broaden the scope of the searches (Efthimiadis, 1996). Plovnick and Zeng (2004) found that the search precision was increased when consumers' lay terms were substituted by preferred terminology from the Unified Medical Language System (UMLS) Metathesaurus. Srinivasan's studies (1996a, 1996b) on the MEDLINE test collection demonstrated the effectiveness of MeSH thesaurus query expansion. Results from studies by Lu et al. (2009) and Stokes, Li, Cavedon,

and Zobel (2009) using the TREC Genomics Track 2006 and 2007 data also confirmed performance improvements after using MeSH thesaurus query expansion. Recently, Yeganova et al. (2009) further explored this issue on the PubMed search system.

Studies have been conducted to explore a variety of other help options for both consumer and professional users in health information seeking tasks. A typical application is the introduction of an intermediary system to provide advice about term selection and query expansion via medical thesauri (Fidel & Efthimiadis, 1995). For example, the Real-Time Query Expansion (RTQE) suggested a list of additional terms for users' query reformulation (White & Marchionini, 2007). A comparative study of RTQE and a baseline interface with no query expansion support concluded that RTQE resulted in better quality initial queries, more engagement in the search, and an increase in the uptake of query expansion. Another example is the Health Information Query Assistant (HIQuA) system, which offers additional UMLS query terms to be used as building blocks for a better, more specific query (Zeng et al., 2006). An evaluation study of the HIQuA system with 213 participants showed statistically significantly higher rates of successful queries as compared to a non-query expansion baseline system.

The structure navigation approach attempts to provide help with a hierarchical MeSH tree structure. Each MeSH term is a node in the tree with a parent node (broader term), a number of children nodes (narrower terms), and some sibling nodes (related terms). With this approach, not only is the definition of the MeSH term given, but also a selection of broader/narrower/related terms. Such a tree structure helps to show the MeSH terms' hierarchical relationships. A well-known structure navigation application is the MeSH Browser provided by the National Library of Medicine (NLM).¹ A comparison study of six MeSH search/navigation interfaces indicated that the MeSH Browser supported concept-oriented navigation with a 63% success rate on mapping natural language patron terms to the MeSH vocabularies (Gault et al., 2002). Despite its helpfulness, a key drawback of the MeSH Browser, however, is that it is a standalone application and is not seamlessly integrated into any health information retrieval system.

In summary, from this brief literature review, we found that many studies have been conducted to explore effective thesaurus-based help mechanisms and to better understand users' search behaviors in health information retrieval. However, no research has been found, to our knowledge, that investigated the explicit integration of MeSH term definitions and structured browsing help components into health information retrieval systems and how these components influence consumers' search behaviors.

2.1. Research goals

To study the related issues in the integration of MeSH thesaurus help components into health information retrieval, we conducted an empirical comparison study. We designed two health information retrieval systems: one had a simple interface to support traditional query-based searching, which we called SimpleMed; and the other, which we called MeshMed, was augmented with two additional components—MeSH Term Browser and MeSH Tree Browser. In this study we have three major research goals:

1. to compare the effectiveness of MeshMed to SimpleMed in supporting consumers' health information retrieval,
2. to examine the roles of the two new MeSH help components implemented in MeshMed towards effective information retrieval, and
3. to explore whether and how the two new MeSH help components influence users' search behaviors and search strategies.

2.2. Methodology

In this section we first introduce the interfaces of the two study systems: SimpleMed and MeshMed. We then describe the methodology of the user study, including data corpus, study tasks, topics, participants, measurements, and study design and procedure.

2.3. System interface

2.3.1. SimpleMed

One health information retrieval system we developed is called SimpleMed. It supports two basic components. The main component is called the Search Browser, a general search component similar to most web search systems such as Google. After a query is given, a list of ranked documents is presented and the words in the query are highlighted. In addition to the document title, part of the abstract and the MeSH terms are also displayed for each retrieved item. A click on a document title will open a new Popup Page component as a tabbed page. In this way, the main search interface (Search Browser) is always available and users do not have to use the "back" button repeatedly to return to their result list.

¹ <http://www.nlm.nih.gov/mesh/MBrowser.html>.

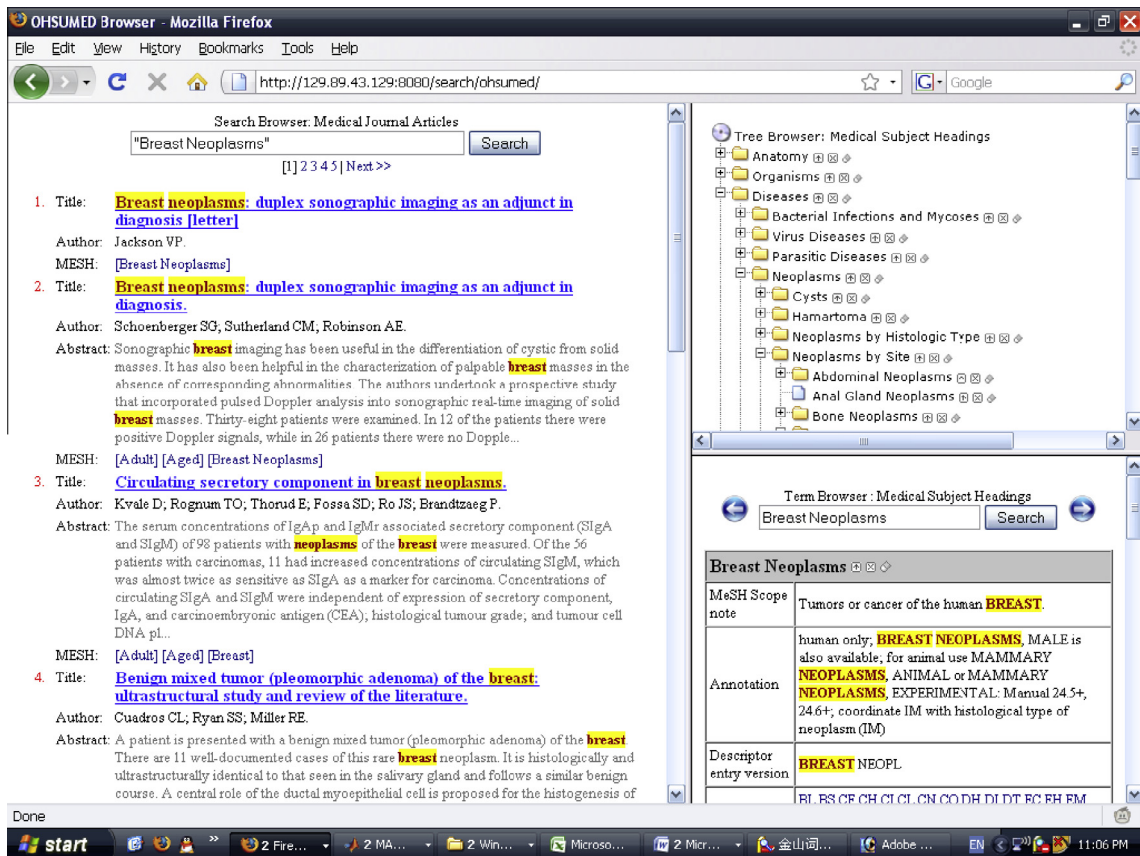


Fig. 1. MeshMed user interface.

2.3.2. MeshMed

The second health information retrieval system we developed is called MeshMed, which supports natural language search, MeSH term search, and hierarchical MeSH tree navigation. MeshMed was implemented based on SimpleMed and augmented with two new components: the MeSH Tree Browser and the MeSH Term Browser. In Fig. 1, on the left is the Search Browser that is available in both systems, to its right on the top is the MeSH Tree Browser, and beneath the MeSH Tree Browser is the MeSH Term Browser.

The MeSH Tree Browser, or Tree Browser for simplicity, displays the hierarchical tree of MeSH terms. It allows a user to expand or collapse a tree node to navigate to more general or more specific MeSH terms. The MeSH Term Browser, or Term Browser, allows a user to search related MeSH terms and their definitions using natural language queries. After the "Search" button is pressed in the Term Browser, a list of related MeSH terms is returned. A click on a MeSH term will display details such as MeSH scope note, annotation, descriptor entry version, allowable qualifiers, print entry, entry, MeSH tree number, and previous indexing (Fig. 2). The meaning of each MeSH description item is predefined and can be found at NLM MeSH definitions.² For example, the "scope note" refers to "free-text narrative giving the scope and meaning of a concept" and the "annotation" refers to "free-text information for indexers and catalogers concerning the use of the descriptor record or qualifier record in PubMed."

The three MeshMed components (Search Browser, Tree Browser, and Term Browser) are not simply displayed together, but are also synchronized with each other. A user's query in the Search Browser will automatically update the Term Browser to list related MeSH terms. In the Tree Browser users can "zoom in" to the hierarchical MeSH tree to focus on the related MeSH terms. A click on a new MeSH term in the Term Browser will trigger an immediate update of the MeSH hierarchical tree in Tree Browser.

Both the Tree Browser and Term Browser are supported with the MeSH vocabulary downloaded from the National Library of Medicine (NLM) in XML format. We parsed the XML data into a backend MySQL database and linked it to our Lemur search engine (Ogilvie & Callan, 2001). The user interface was implemented in PHP and we developed a PHP-Java adaptor to connect the Java search engine to support real-time search.

² http://www.nlm.nih.gov/mesh/xml_data_elements.html.

Breast Neoplasms	
MeSH Scope note	Tumors or cancer of the human BREAST .
Annotation	human only; BREAST NEOPLASMS , MALE is also available; for animal use MAMMARY NEOPLASMS, ANIMAL or MAMMARY NEOPLASMS, EXPERIMENTAL: Manual 24.5+, 24.6+; coordinate IM with histological type of neoplasm (IM)
Descriptor entry version	BREAST NEOPL
Allowable Qualifiers	BL BS CF CH CI CL CN CO DH DI DT EC EH EM EN EP ET GE HI IM ME MI MO NU PA PC PP PS PX RA RH RI RT SC SE SU TH UL UR US VE VI
PRINT ENTRY	Breast Cancer
PRINT ENTRY	Breast Tumors
PRINT ENTRY	Cancer of Breast
ENTRY	Breast Neoplasm
ENTRY	Breast Tumor

Fig. 2. Details in the term browser.

Table 1
Descriptions of the six search topics.

Topic ID	Search topic description	OHSUMED ID
1	You are a 50 year old person with COPD	84
2	You want to know how theophylline should be used for chronic and acute asthma	
	You are a patient with cerebral palsy and depression	20
	You want to know about the relationship between cerebral palsy and depression	
3	You have diabetic gastroparesis	38
	You want to know what the best treatment is for diabetic gastroparesis	
4	You are a patient with a migraine	63
	You want to know about the treatment of migraine headaches with beta blockers and calcium channel blockers	
5	You are a 60 year old menopausal woman without hormone replacement therapy	1
	You want to know if there are adverse effects on lipids when progesterone is given with estrogen replacement therapy	
6	You are a 60 year old man with severe malabsorption	95
	You want to know about the processes of infiltrative small bowel and information about small bowel lymphoma and heavy alpha chain disease	

2.4. Study participants

We recruited study participants from all across the campus of a 4-year university. Flyers were posted on campus and a total of 55 participants were recruited in the order of their responses. All participants were currently enrolled undergraduate or graduate students. After the data collection, 45 data sets were valid. We discarded 10 data sets due to a pilot study, technical problems in data recording (nine) and other reasons (a participant did not finish). Each participant was paid either \$15 or \$10 as a token of appreciation.

2.5. Data corpus

The data corpus used for our usability study is the OHSUMED³ test collection (Hersh, Hickam, Haynes, & McKibbin, 1994). OHSUMED is a clinically-oriented MEDLINE subset, consisting of 348,566 citations covering every reference from 270 medical journals over a 5-year period (1987–1991). The fields in each citation include title, abstract, MeSH term list, author, source, and publication type. Also included in the collection are 106 search topics.

We randomly selected six search topics from the 106 topics for the user study (Table 1). These tasks need users to explore related documents to better answer the questions. The search topics were created based on the original OHSUMED queries. For an example, one query description in OHSUMED is

³ <http://ir.ohsu.edu/ohsumed/>.

Patient information: 60 year old menopausal woman without hormone replacement therapy

Information request: Are there adverse effects on lipids when progesterone is given with estrogen replacement therapy

Our search topic is phrased as

You are a 60 year old menopausal woman without hormone replacement therapy. You want to know if there are adverse effects on lipids when progesterone is given with estrogen replacement therapy.

2.6. Search tasks

Search task is an important variable that influences the retrieval performance and users' search behavior (Kim & Allen, 2002; Tang, 2007). In this study we had two types of tasks. One was to provide a definition of a key biomedical term selected from each search topic. For example, "What are calcium channel blockers?" There was one definition question for each search topic and there were six questions in total (see Appendix A). The other was to find the relationships between biomedical concepts and to provide an answer for each search topic (as described in Table 1). Participants were not allowed access to the search systems while answering questions.

2.7. Measurements

To address our first research goal about system effectiveness, we measured participants' perceived topic familiarity change, time spent and question-answer performance. The perceived topic familiarity change score was obtained from the differences of reported topic familiarity before and after each search. Similar to the approaches in previous studies (Kelly & Cool, 2002; Shiri & Revie, 2003), we used a nine-point Likert scale to assess participants' perceived topic familiarity. The time spent refers to how much time a participant spent interacting with a search system. We believe this variable partly reflects participants' engagement in search tasks, which should be encouraged in a successful browsing-based search tool (O'Brien and Toms, 2008; White & Marchionini, 2007).

A more objective way to assess the search effectiveness is a user's ability to complete a search task (Coiera & Vickland, 2008). The third variable towards the effectiveness evaluation relates to participants' question answer performance, which was computed based on the scores participants answering the questions. Each answer was graded as wrong, partially correct, and correct. The grading guidelines were developed by a panel of four people: two usability study specialists and two domain specialists (one is a clinical doctor and another is a medical school graduate student). The grading was then completed independently by the two domain specialists. The Weighted Cohen Kappa value for inter-rater reliability was high ($\kappa = 0.696$). The disagreements between the two sets were solved in the panel discussion and by consulting additional outside experts. For validation purposes, we further conducted a Spearman's rho correlation analysis between participants' question answer performance scores and the reported topic familiarity change scores.

In accordance to the three variables, we have three hypotheses:

- H1: users perceive that they gain more topic familiarity when using MeshMed as opposed to SimpleMed.
- H2: users are more likely to answer the questions correctly when using MeshMed as opposed to SimpleMed.
- H3: users are encouraged to spend more time interacting with and searching the system when using MeshMed as opposed to SimpleMed.

For simplicity, in the remainder of the paper we just use the phrase "topic familiarity" to refer to the perceived topic familiarity. In H2, the correct answers to which we refer are correct or partially correct answers.

To examine the contributions of the two new components towards effective health information retrieval, we used another two variables. One is the participant reported contributions, in percentages, by each component towards their topic familiarity change. Another is the time a participant spent on each system component.

Considering the subjective nature of the participant's reported component contributions, we also supplemented our findings with participants' search pattern analysis and post-experiment comments.

To explore users' search behaviors and search strategies, we coded participants' search patterns based on the transaction log data that recorded each participant's actions during the search process (i.e., clicking on a component, giving a query, closing a component). According to the number of participants' clicks on the search system components and the sequences of these clicks, we extracted the pattern of shifts between different search system components and the frequencies of using each component. Typical search strategies were then summarized from analyzing these coded patterns and shifts.

Finally, we triangulated our findings with participant comments gathered from the post-experiment questionnaire.

2.8. Study design and procedure

A within-subject design was used. Each participant used both MeshMed and SimpleMed to search six topics (three with MeshMed and three with SimpleMed). We used 6×6 Latin Square design to balance the learning effect on different search topics and the two interfaces (Raghavarao & Padgett, 2005).

Even though in this study we focus on the main effect of the interface, it is noted that such an effect may differ between search topics. Due to our randomization design, each subject answers different questions in each interface. This design limited the ability to test interaction effect between the topics and interfaces, which may lead to the ignorance of the difference of the interface effect for different topics. However, each search topic is equally represented in both interfaces in our randomization design and we believed that our investigation on interface main effect is still valid.

A 10 min tutorial by one of the researchers was given to each participant to demonstrate the major functions of the key components in both systems. During our demonstration, we did not lead the participants to believe one system was superior to the other. Using neutral language, we explained that we wanted to understand people's search behaviors and to examine the factors that influence the effectiveness of health information retrieval. After the tutorial, the participants were required to sign a consent form for the study and fill out a pre-experiment questionnaire asking for their gender, major, experience using the Internet, and how often they search for health information on the Internet. In the study, each participant used one system (MeshMed or SimpleMed) to search three topics, and then shifted to another one to finish the remaining three topics. To maintain neutrality, we labeled the systems as System A and System B, and balanced the order of the systems that each participant worked with to avoid bias. There was no timeframe limitation. Participants decided when they should finish searching and answer questions.

Before beginning a search on a topic, participants reported their perceived familiarity with the topic using a nine-point Likert scale. The same scale was given immediately after the participant finished the topic search in order to record the participant's topic familiarity change. The participants were also asked to specify how much each component (Search Browser, Term Browser, Tree Browser, and Popup Page) contributed to their topic familiarity changes.

After participants finished all six search topics, an overall post-experiment questionnaire was given. It asked for information about: (1) which interface they preferred and why, (2) the overall contribution of the four components to their perceived topic familiarity changes, (3) their comments on each of the four components, and (4) suggestions and other comments.

The participants' interactions with the system were automatically recorded and stored in a MySQL database as part of the transaction log data. These interactions included: query formulation in the Search Browser, document opening in the Popup Page, clicking in the Tree and Term Browser, searching in the Term Browser, and opening/collapsing a tree node in the Tree Browser.

3. Results and analysis

Among the 45 participants, 16 of them were male (35.56%) and 29 were female (64.44%). Most – 44 out of 45 – participants searched the Internet daily (97.8%). Regarding their history of searching for health information on the Internet, three (6.67%) reported “daily,” seven (15.6%) reported “weekly,” 30 (66.7%) reported “occasionally,” and five (11.11%) reported “never.” These 45 participants came from 12 different academic majors: Architecture and Urban design, Business, Civil Engineering, Nursing, Computer Science, Conservation and Environmental Science, Electrical Engineering, Geography, Information Studies, Mechanical Engineering, Occupational Therapy, and Global Studies.

3.1. Search effectiveness

3.1.1. Perceived topic familiarity change

Table 2 lists the perceived topic familiarity (before and after the search) and the topic familiarity change for each of the six topics for both the SimpleMed and MeshMed interface. We noted that except for topic one, MeshMed was reported to be more helpful for increasing participants' topic familiarity. On average, participants reported a greater topic familiarity

Table 2
Participants' perceived topic familiarity change for the SimpleMed and MeshMed interface, respectively.

Topic ID	Before	After	Change (percentage)
<i>SimpleMed</i>			
1	1.85	4.80	2.95(159.46%)
2	2.32	2.92	0.60(25.86%)
3	1.60	3.52	1.92(120.00%)
4	2.77	4.36	1.59(57.40%)
5	2.13	4.13	2.00(93.90%)
6	1.30	2.80	1.50(115.38%)
Average	2.00	3.76	1.76(88.00%)
<i>MeshMed</i>			
1	2.16	4.72	2.56(118.52%)
2	2.25	4.50	2.25(100.00%)
3	1.85	5.15	3.3(178.38%)
4	2.52	4.61	2.09(82.94%)
5	2.41	5.18	2.77(114.94%)
6	1.40	3.32	1.92(137.14%)
Average	2.10	4.58	2.48(118.10%)

Table 3

Distribution of question–answers in the three grading categories (Wrong, Partly Correct, Correct).

	Wrong	Partly Correct	Correct	Total
<i>Simple Med</i>				
Count	27	29	45	101
Percentage	26.7	28.7	44.6	100.0
<i>MeshMed</i>				
Count	11	43	69	123
Percentage	8.9	35.0	56.1	100.0
<i>Total</i>				
Count	38	72	114	224
Percentage	17.0	32.1	50.9	100.0

Table 4

Time spent (in seconds) on the four components: Search Browser, Popup Page, Term Browser, and Tree Browser.

Topic ID	Search browser	Popup page	Term browser	Tree browser	Total
<i>SimpleMed</i>					
1	97.86	157.33			255.19
2	231.80	106.60			338.40
3	90.48	181.92			272.40
4	96.05	239.45			335.50
5	82.65	211.74			294.39
6	178.15	132.50			310.65
Average	129.50	171.59			301.09
<i>MeshMed</i>					
1	116.32	113.40	88.20	19.56	337.48
2	132.10	63.45	106.80	35.85	338.20
3	58.75	77.45	95.90	20.60	252.70
4	133.91	109.74	131.91	33.09	408.65
5	60.30	150.78	79.96	24.70	315.74
6	103.40	71.92	140.40	42.80	358.52
Average	100.80	97.79	107.195	29.43	335.22

change (2.48 or 118.10% increase) when using MeshMed compared to using SimpleMed (1.76 or 88.0% increase). This is particularly true for topic two (100% vs. 25.86%).

With the randomization design, it was assumed that the two user groups (MeshMed vs. SimpleMed) would be homogenous in terms of previous familiarity of the topic. The data supports this assumption given the mean and standard deviation of the two groups' pre-search topic familiarity scores. The mean topic familiarity score for SimpleMed was 2.01 with a standard deviation 1.43. The mean topic familiarity score for MeshMed was 2.08 with a standard deviation 1.60.

In order to compare the mean topic familiarity change scores of both groups, we first averaged the topic familiarity change scores for three questions in each interface for each subject, and then applied a two-tailed paired sample *T*-test (we have tested that the average topic familiarity change scores in two groups follow the normal distribution). The result showed a significant difference between the two groups ($p = 0.002 < 0.01$, $df = 44$, $t = -3.295 [-1.197, -0.28836]$). This indicated that the MeshMed group reported a significantly higher perceived topic familiarity change score than the SimpleMed group (Hypotheses H1).

3.1.2. Performance testing scores

There were 45 participants in the study and each answered six questions. As a result, we obtained 270 question–answers. Among them, 224 were validated answers (83.0%). The remaining 46 question–answers (17.0%) were either blank or “N/A.” We did not count them as wrong answers because in the study we allowed participants to skip questions. There were 34 such answers from participants using SimpleMed and 12 from participants using MeshMed. For the 224 validated answers, 101 of them were from participants using SimpleMed (45.09%) and 123 from participants using MeshMed (54.91%).

Table 3 lists the distribution of the question–answers in the three grading categories (wrong, partly correct, correct) for both SimpleMed and MeshMed. We noted that more wrong answers (26.7%) were given when using SimpleMed for searching than using MeshMed (8.9%). Meanwhile, more correct answers were given with MeshMed (56.1%) than with SimpleMed (44.6%). A Chi-Square association test indicated that there was a significant association between interface type and answer test scores ($\chi^2 = 12.471$, $df = 2$, $p = 0.002 < 0.01$). This result confirmed that MeshMed users were more likely to answer the questions correctly or partially correctly than the SimpleMed users (Hypotheses H2).

Table 5

Contribution of each component (Search Browser, Popup Page, Term Browser, and Tree Browser) towards the participants' topic familiarity change.

	T1 (%)	T2 (%)	T3 (%)	T4 (%)	T5 (%)	T6 (%)	Overall (%)	Average (%)
<i>SimpleMed</i>								
Search	43.69	48.09	55.72	41.95	45.74	52.35	47.62	47.92
Popup	56.31	51.91	44.28	58.05	54.26	47.65	52.38	52.08
<i>MeshMed</i>								
Search	29.91	20.07	28.03	27.75	27.25	21.76	26.11	25.79
Tree	12.85	21.15	15.61	13.35	14.05	15.98	15.17	15.50
Term	35.75	36.20	33.12	38.48	29.50	38.02	35.25	35.18
Popup	21.50	22.58	23.25	20.42	29.21	24.24	23.47	23.53

Table 6

Percentage of time spent on Search Browser, Popup Page, Term Browser, and Tree Browser.

	T1 (%)	T2 (%)	T3 (%)	T4 (%)	T5 (%)	T6 (%)	Average (%)
<i>SimpleMed</i>							
Search	38.35	68.65	33.22	28.63	28.08	57.35	42.35
Popup	61.65	31.50	66.78	71.37	71.92	42.65	57.65
<i>MeshMed</i>							
Search	34.47	39.06	23.25	32.77	19.10	28.84	29.58
Tree	5.80	10.60	8.15	8.10	7.82	11.94	8.73
Term	26.13	31.58	37.95	32.28	25.32	39.16	32.07
Popup	33.60	18.76	30.65	26.85	47.75	20.06	29.61

We further conducted a Spearman's rho correlation analysis between the question–answer scores and the reported topic familiarity change scores. The results indicated a 0.213 positive correlation ($N = 224$) and it was statistically significant ($\alpha = 0.001 < 0.01$). In other words, the greater the participants reported their topic familiarity change, the higher they achieved in their question–answer scores.

3.1.3. Time spent

Table 4 lists the time spent by participants on MeshMed and SimpleMed, as well as the specific time spent on the components. In order to compare the mean total time spending of both groups, we first averaged the spending time for three questions in each interface for each subject, and then applied a two-tailed paired sample T-test (After square root transformation, the average time spent in two groups follows normal distribution). The result did not show a statistically significant difference between the two groups ($p = 0.185 > 0.05$, $df = 44$, $t = -1.346 [-2.216, 0.441]$) and thus the Hypotheses H3 was rejected.

In summary, we found MeshMed was more effective than SimpleMed in terms of the topic familiarity change and the question–answer performance. But we did not find a statistically significant difference regarding the time users spent on the two systems.

3.2. Component contributions

In the study, we also asked participants to report the contribution of a component (Search, Term and Tree Browsers and Popup Page) towards their topic familiarity change for each search topic. We normalized these scores so the total contribution from each component would add up to 100% and the results are listed in Table 5.

We noted that even though the reported contribution of each component changed from topic to topic, the percentage value of the overall contribution reported for the six topics (“Overall” column in Table 5), which was given by participants when they finished all search tasks, was almost identical to the percentage value of the average contributions (“Average” column), which was computed based on the contributions given by participants after each search topic (an average of T1 through T6 in Table 5). This finding confirmed, to some extent, the reliability and consistency of the user reported contribution of each component.

On average, we found that the contribution from the Search Browser was close to the contribution from the Popup Page for the traditional SimpleMed interface. When rating MeshMed, users reported that the most useful component was the Term Browser (35.18%), contributing 34.2% more than the traditional Search Browser component (25.79%). Relatively, the Tree Browser contributed the least towards a participant's topic familiarity change (15.5%).

Table 6 lists the percentage of time spent on each of the components for the two systems. On average, for SimpleMed, users spent more time on the Popup Page (57.65%) than on the Search Browser (42.35%). With MeshMed, the Term Browser was the most widely used component (32.07%), while the Tree Browser was the least used one (8.73%).

Table 7

Frequency of component shift when using the MeshMed interface.

	Component shift	Frequency	Percentage
1	Search Browser → Popup	118	18.88
2	Term Browser → Search Browser	117	18.72
3	Search Browser → Term Browser	113	18.08
4	Popup → Search Browser	61	9.76
5	Pop-up → Term Browser	49	7.84
6	Tree Browser → Term Browser	34	5.44
7	Tree Browser → Search Browser	32	5.12
8	Search Browser → Tree Browser	30	4.80
9	Term Browser → Popup	29	4.64
10	Term Browser → Tree Browser	26	4.16
11	Popup → Tree Browser	8	1.28
12	Tree Browser → Popup	8	1.28

Table 8

Shifts to other MeshMed components in number and percentage.

From/To	Search browser	Popup page	Term browser	Tree browser	Total
Search browser	–	118(45.21%)	113(43.30%)	30(11.49%)	261
Popup page	61(51.69%)	–	49(41.53%)	8(6.78%)	118
Term browser	117(68.02%)	29(16.86%)	–	26(15.12)	172
Tree browser	32(43.24%)	8(10.81%)	34(45.95%)	–	74

Comparing the data of Tables 5 and 6, we noted that the distribution patterns were similar. In general, the more time participants spent using a component, the more helpful they reported that component. Yet, there was one instance in which the result diverged from this general rule. Participants spent only 8.73% of their time using the Tree Browser but perceived its contribution as relatively high (15.5%). In general, participants spent more time on the Term Browser and considered it as the most helpful component. We also noted that participants regarded the Tree Browser, another new component, as the least helpful component towards their topic familiarity changes, and did not use it as much as other components. However, the two search components did not cancel each other out. While the Tree Browser on its own did not demonstrate a greater contribution to users' topic familiarity changes than the Search Browser and the Popup Page, neither new search component provided a negative contribution to users' topic familiarity changes. Data obtained from the study questionnaire, which will be discussed later, confirmed that users benefited most by using a combination of different search components resulting in rich search strategies.

3.3. Patterns and behaviors

3.3.1. Between-component interactions

For the traditional SimpleMed system, users' search strategies are limited because there are only two between-component shifting actions available: from the Search Browser to the Popup Page or vice versa. For the MeshMed interface, however, users may have additional options to choose from. Given four functional components in MeshMed, there are 12 unique between-component shifts that are possible. Based on our study transaction log data, Table 7 lists these between-component shifting patterns founded and their frequency of use. We noted that the interactions between the Term Browser and the Search Browser (No. 2 and No. 3 in frequency) were almost as frequent as the traditional move from the Search Browser to the Popup Page pattern. In the study, we observed that many participants used such Search Browser–Term Browser interaction to help form and reform their queries. After a trial in the Search Browser, participants searched unfamiliar medical terms in the Term Browser, and then reformed the queries in the Search Browser in order to improve their search. Such an interaction pattern, combining with the Search Browser–Popup Page iterative interaction, could be helpful for a consumer to establish a clear concept about the search topic. We will further discuss this issue in Section 4.

Table 8 provides an alternate way to look at the patterns of component shifts and provides us with a number of interesting observations. For example, the table shows that there were 74 shifts from the Tree Browser to some other component. Of these 74, 43.24% shifted to the Search Browser, 10.81% shifted to the Popup Page, and 45.95% shifted to the Term Browser. A possible scenario is that, after navigating through the Tree structure, users employed their newly updated topic concept in the Search Browser to clarify, verify, or modify the queries, or look for new term definitions in the Term Browser. Another noteworthy shift occurred after participants used the Term Browser. We noted that the majority of shifts (68.02%) were to the Search Browser. This observation may indicate that the Term Browser worked as a tool for participants to bridge the medical vocabulary gap so more accurate queries could be formulated in the Search Browser.

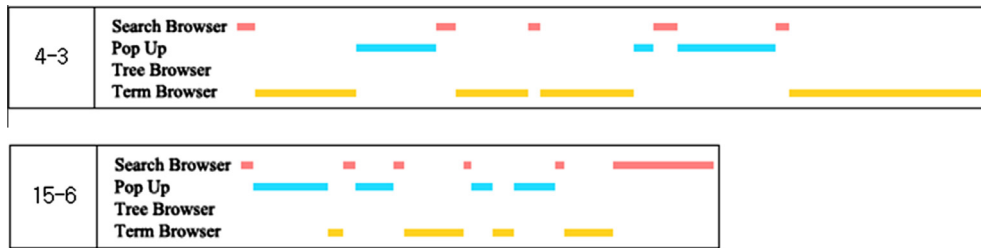


Fig. 3. Example of user search behavior. It illustrates users' time spending on different component along the time line while they do search. The first column indicates user ID and the topic they are currently searching (4-3 means user 4 search topic 3). The length of the bar in the second column indicates how much time the user spent on a component (Search Browser, Pop Up, Tree Browser, and Term Browser) during their search.

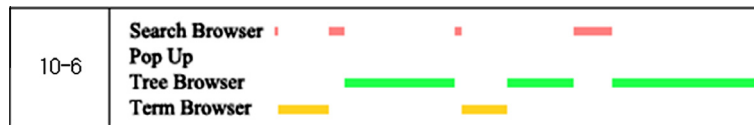


Fig. 4. Example of user search behavior.

3.3.2. Typical search strategies

By using different between-component shifts and their combinations, a participant was able to develop a unique search strategy to better fit his/her search experience and skills. After analyzing the topic searches conducted in MeshMed for all 45 participants (with a total of 135 individual search cases), we identified seven typical search strategy patterns.

In the list that follows, we summarize each of these patterns with a short name, a description, a list of examples, and a brief analysis. Each example is coded with a user ID and search topic ID. For example, "4-3" means "user 4 searches topic 3." Visual charts are provided for some of the examples. Each chart represents one search case and has two columns. The first column is the case number (e.g. 4-3). The second column visually displays the search process in terms of the time spent using each MeshMed component (Search Browser, Popup, Tree Browser, and Term Browser). For example, a bar aligned with the Search Browser indicates that the user is actively using the Search Browser for searching, navigating, or query inputting. The length of the bar represents the time spent on that component before shifting to the next component or ending the search. All of these bars are drawn along the time line from the left to the right covering the entire search for that case. In case 4-3, for example, user 4 is searching topic 3 (see below). After beginning her search with the Search Browser, she used the Term Browser, then the Pop up Page, and so on, until she finished her search with the Term Browser.

1. Search Browser/Term Browser/Popup Page Pattern:

Description: This is the most commonly used search pattern we observed. In addition to the familiar Search Browser–Popup Page interaction, the Term Browser was employed to search definitions of unfamiliar medical terms.

Examples: 4-3, 15-6, 7-5, 20-3

Case analysis:

- Comments from the users confirmed the contribution of the Term Browser in this pattern: "When I wanted a definition this [the Term Browser] was the most helpful" (user 4). "It [the Term Browser] provides a quick definition for topics, and allows you to replace or add search terms to the search browser" (user 15) (see Fig. 3).

2. Search Browser/Term Browser/Tree Browser Pattern:

Description: Interactions between the Search Browser, Term Browser, and Tree Browser

Examples: 10-6

Case analysis:

- User 10 frequently chose the Search Browser, Tree Browser, and Term Browser. She stated: "I liked having a combination of all three aspects to search. I was able to build my knowledge with the different pieces each browsing method offered. Without the Tree and Term browser I had difficulty establishing a base knowledge of the general concepts" (see Fig. 4).

3. Tree Browser/Term Browser Pattern:

Description: Participants moved either from the Tree Browser to the Term Browser or from the Term Browser to the Tree Browser. The pattern could be applied at the beginning of a topic search or in the middle.

Examples: 3-6, 8-4

Case analysis:

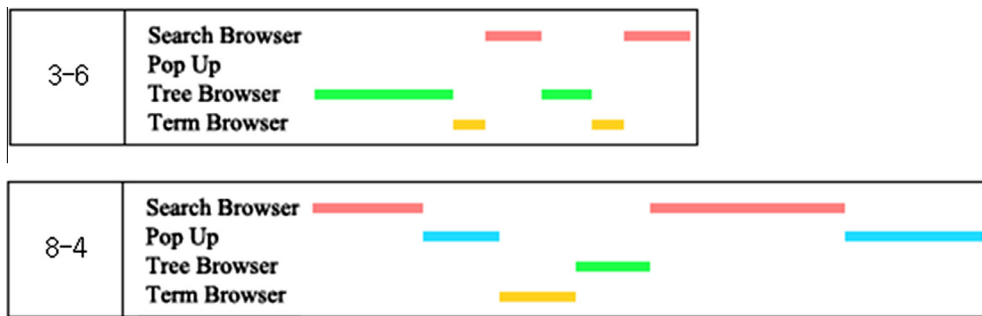


Fig. 5. Example of user search behavior.



Fig. 6. Example of user search behavior.

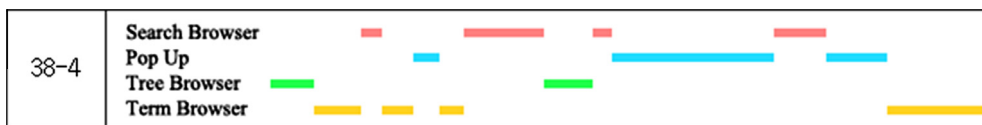


Fig. 7. Example of user search behavior.

- In case 3-6, the participant spent a significant amount of time using the Tree Browser in the beginning. She commented that “When you know what you are looking for, it [the Tree Browser] helps you to narrow down your search very quickly.”
- In addition to the traditional Search Browser/Popup Page pattern, the user in case 8-4 applied the Tree/Term Browser combination to further understand the meanings of some medical terms. This user stated, “The term browser allows the user to focus on key words and phrases – gaining a general understanding before moving onto more complicated concepts/groups of concepts” (user 8) (see Fig. 5).

4. Tree Browser/Search Browser Pattern:

Description: Intensive interactions between the Search Browser and the Tree Browser.

Examples: 24–2

Case analysis:

- This is the participant's third MeshMed topic. She had used the Tree Browser to search for information in the prior two topics. She felt that she “needed to be familiar with the topic to use the hierarchy” (user 24) (see Fig. 6).

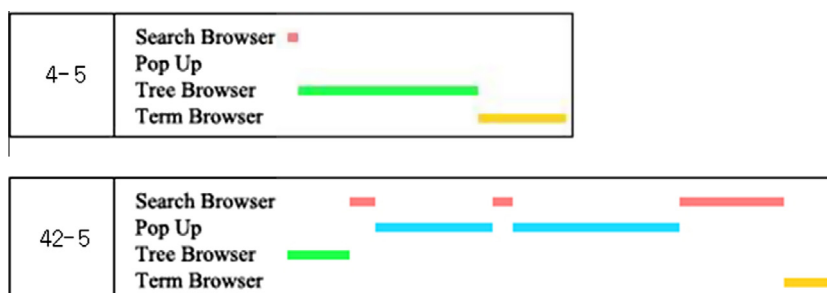


Fig. 8. Example of user search behavior.

5. Holistic pattern:

Description: Users employed a combination of different components and shifted between them frequently.

Examples: 38–4

Case analysis:

- In case 38–4, we observed a balanced use of all four components. User 38 commented that “the combination of the three approaches was helpful” (see Fig. 7).

There are two additional patterns that are related to the Tree Browser.

1. Tree-first Pattern:

Description: Participants began with the Tree Browser and then shifted to other patterns.

Examples: 42–5, 4–5

Case analysis:

- User 4 explored the use of the Tree Browser at the beginning of her first MeshMed topic. She did not explicitly use the Tree Browser in the subsequent two MeshMed topics but she indicated that she liked the idea of starting a search with the Tree Browser. She said, “This [using the Tree Browser] was a good start to the search. It allowed me to view terms without having to input anything.”
- User 42 gave a negative comment regarding the Tree Browser, stating: “I didn’t like it. [There are] Too many options. [It is] Kind of confusing and cluttered” (see Fig. 8).

2. Tree for query terms pattern:

Description: Users selected the Tree Browser to add broader/narrower/related terms to their search queries.

Examples: 15–2, 11–6, 18–3

Case analysis:

- In case 11–6, the user did not explicitly click on the Tree Browser. However, after typing “Heavy Chain Disease” in the Term Browser, the Tree Browser was updated simultaneously with a narrower term (“immunoproliferative small intestinal disease”). User 11 then entered the narrower term from the Tree Browser into the Search Browser.
- In case 18–3, the participant used the Tree Browser to understand the term “Domperridome” (see Fig. 9).

3.4. Participants’ comments

After finishing all six search topics, the participants were asked to share their comments about each component.

3.4.1. The term browser

The majority of participants (34 out of 45) explicitly mentioned that the Term Browser was helpful in providing medical term definitions. For example:

“Very effective...simple point and click interface, quick access to accurate, concise information.” (user 13)

“The term browser is best for finding definitions of the medical terms. Since the articles are geared towards medical professionals this is very helpful.” (user 23)

Some participants also mentioned that the Term Browser was easy to use and allowed them to “replace or add search terms to the search browser” (user 15).

3.4.2. The tree browser

Most users (27 out of 45) explicitly indicated that the Tree Browser gave them a structured view of the related terms. For example:

“You can see what general category things belong to, and if you are not finding anything in your search it can help you find related keywords.” (user 15)

Several users, however, commented on negative qualities of the Tree Browser. For instance, one participant stated that it was “too wide/general, did not offer me any sort of specific information, really” (user 13). Such negative experiences might be associated with participants’ inappropriate use of the Tree Browser. We will further analyze this issue in the discussion section.

3.4.3. MeshMed vs. SimpleMed interface

In the post-experiment questionnaire we asked each participant to specify which interface (MeshMed or SimpleMed) they preferred and why. Nearly all participants (43 out of 45) chose MeshMed. The primary merit identified by the participants was that MeshMed provided them with alternative means to search for information. For example:

“Having multiple options for browsing allows the user to use multiple strategies to locate the desired information.” (user 8)

Nine participants indicated that the combination of the Tree Browser and the Term Browser matters. For instance:

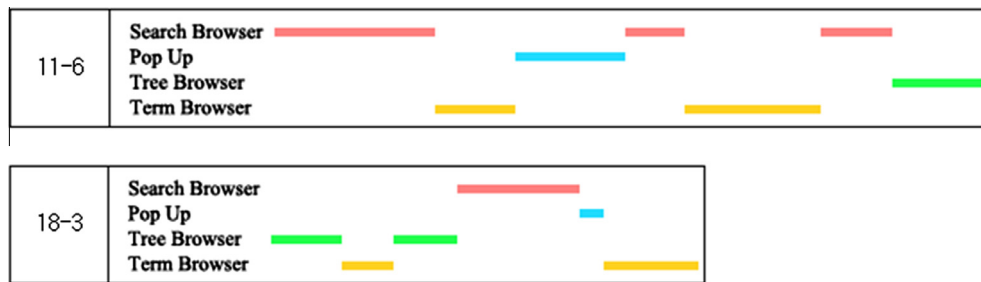


Fig. 9. Example of user search behavior.

"I think it's helpful to have a definition of the terms and the tree browser to display the hierarchical structure." (user 11)

"It was easier to use the term search and the tree search for subjects that may be unfamiliar." (user 18)

Before the study we were uncertain if adding two additional components (Term Browser and Tree Browser) to the MeshMed interface would overwhelm the users. In the post-experiment questionnaire, however, we did not find any complaints regarding this issue. In fact, three participants specifically mentioned that MeshMed was easier to use. For example:

"I received better search results. I also found it easier to use" (user 11).

4. Discussion

4.1. New MeSH components and search strategies

Our comparison of SimpleMed and MeshMed found that although adding two new components took away from a simplified interface, users preferred MeshMed when researching health information. MeshMed was more effective than SimpleMed in terms of increasing topic familiarity and boosting question-answering performance. Despite the clean and simple interface of SimpleMed, most participants considered MeshMed to be more helpful.

Our findings correspond well to Wildemuth, (2004) conclusion that the level of domain knowledge affects search patterns. In Wildemuth's study, when students' domain knowledge was low, moves and interactions with the search systems were high. At this stage users often added additional concepts to achieve the desired results. We believe that the two new MeSH components were particularly helpful in terms of new concept construction and additional concept expansions. By providing two new MeSH components, MeshMed not only supports traditional natural language queries, but also allows consumers to look for medical term definitions and navigate the hierarchical medical term tree structure. Some comments from the participants confirmed the effectiveness of the Term Browser in concept construction.

"It [Term Browser] gave quick and understandable definitions and relations to other terms. Very helpful in determining what the original concept was." (user 10)

"I LOVED the MeSH scope notes (in the Term Browser). The quick definitions greatly helped in my understanding of the topic, and it was very helpful when related terms were suggested – because the main search page was right there, I could just go ahead and enter the new terms in a key word search. This interface made the scientific documents much less intimidating." (user 28)

The main benefit of adding the Term Browser and the Tree Browser is to enable users to develop diversified search strategies. The Term Browser directly provides MeSH term definitions and descriptions; while the Tree Browser enables users to further explore the knowledge structure related to a new concept. The rich component interaction patterns presented in our study, along with the positive comments from study participants, provided evidence about the effectiveness and usefulness of the new augmented MeshMed system.

Another interesting observation, which was also corresponding well to Wildemuth's finding (2004), was the participants' search strategy changes. We found that participants did not always stick to the same search strategy. Based on the observations and transaction log analysis, we noted that, at the beginning, many participants attempted to use the Tree Browser to locate a term and to find clues for an effective query. Yet, without possessing the necessary domain knowledge about the related medical topic, it was like trying to find a tiny street name on a large, unfamiliar city map. Their frustration came from using the map inappropriately. Some negative comments from participants about the Tree Browser may be attributed to using the Tree Browser in this way.

As users became more familiar with the MeshMed interface, their search strategies also changed. One of the main advantages of MeshMed is the synchronized interaction between the different components. After a query is submitted in the Search Browser or the Term Browser, the Tree Browser will automatically zoom into related MeSH terms and tree structure. As a result, broader/narrower/related terms will be visually displayed in a hierarchical manner. This interactive function suggests a better strategy to use the Tree Browser. Let's use the city map analogy again. Once the user finds the street name, the

map becomes very helpful in providing additional information about the neighborhoods such as the location of shopping malls, parks and schools. The Tree Browser has its unique contribution of presenting related MeSH terms in a structured manner. The “Tree for terms” search strategy (e.g., user 11 on topic 6) we observed provided a good example for such a strategy. We found that many participants began to use this strategy when they became more familiar with the MeshMed interface (usually during the second or third MeshMed topic search). A participant’s comments further explicitly illustrated such strategy changes.

“It was helpful to see how the term was related to other terms. I liked looking at the tree browser after using the search browser to get a broader perspective of the term.” (user 11)

4.2. Cognitive schema perspective

Another way to look at our findings is from the cognitive schema perspective. In psychology, a schema represents a set of rules or scripts used by people to interpret the world (Brewer & Treyns, 1981). People’s cognitive schemas can be roughly classified into three levels: terminology, relationship, and knowledge schema. On the terminology schema level, a person builds concepts using definitions. This is the base upon which the other two types of schemas are formed (Fidel & Efthimiadis, 1995). On the relationship schema level, connections are created between related concepts. This requires deeper cognitive development than the terminology schema since it involves multiple concepts. On the knowledge schema level, people integrate definitions, concepts, and relationships into a well organized knowledge structure. Our study suggests that MeshMed is particularly helpful for users on the concept level. The Term Browser allows a user to directly find definitions, annotations, and related terms, while the Tree Browser helps to present a list of broader, narrower, or related terms in a hierarchical manner. Vakkari’s analytical study (2002) suggested that the concept interrelations inferred from the query terms contribute to the meaning of the conceptual construct which the query represents, and should be reflected in the terms used to expand the initial query. The Term Browser and Tree Browser provided tools to help users gain deep understanding of related terminologies, reformulate queries, and thus develop new search strategies.

4.3. Limitation analysis and future studies

We noticed several limitations for our study. First, about the choosing of baseline system, similar to many previous studies (i.e., White & Marchionini, 2007), we did not use a product health information retrieval system such as PubMed for comparison. Our focus of the study was to explore the roles and contributions of two new MeSH thesaurus components – MeSH Term Browser and MeSH Tree Browser, not to develop a superior system better than PubMed. For this purpose we decided to design and develop two new dedicated prototype systems so we can control other variables in the study (i.e., interface design, ranking algorithms, help mechanisms, and additional functions such as query expansions). We acknowledged that PubMed is treated as the “golden standard” of health information retrieval system. After we have achieved a better understanding about the roles and contributions of some key components in a health information retrieval system, it would be beneficial to further develop more comprehensive applications and compare their performance with PubMed or Google.

Google also indexes the MEDLINE database and its simple interface attracts many consumers (Steinbrook, 2006). Similar to PubMed, Google does not provide immediately available support tools to help users with medical terminology. In this study, Google was not our focus because it is not a search system dedicated to medical information. The quality and accuracy of the health information retrieved cannot be controlled. In spite of its popularity, Google is not recommended by Consumer and Patient Health Information Section (CAHIS) of the Medical Library Association (MLA) as the top-rated health information websites.

Secondly, about the evaluation variables, we used the participants’ topic familiarity changes to partly reflect the search effectiveness. In the study questionnaire, however, we incorrectly referred to topic familiarity as topic knowledge. Even though we verbally explained to study participants that they should report their topic familiarity before and after each search, the inconsistent use of terminology might introduce unnecessary confusion. In hindsight, the topic familiarity is more appropriate and reflects the meaning of this measurement, and should be consistently used in our study.

Another limitation is a potential problem with the method chosen to assess the time participants spent interacting with the system. System recorded user actions (such as query formulation, document opening, and clicking) provided a general overview of participants’ overt physical operations, yet they neglected to include more subtle interactions with the system that did not involve physical movement. For instance, since three of the components (Search Browser, Term Browser, and Tree Browser) are always available on the screen, it is possible that participants were only physically interacting with one component (such as browsing a result list in the Search Browser) but gaining information visually from the other two. In the future, an eye-tracking study like we did in our other studies (i.e., Marchionini & Mu, 2003) would help us more accurately measure this variable.

Finally, an interesting but not studied issue is how the MeshMed interface might influence a user’s cognitive load. By adding two unfamiliar search components, the MeshMed may potentially increase a user’s cognitive load due to its complex interface. On the other hand, MeshMed might also help reduce a user’s cognitive load because it provides more search strategy options. Furthermore, the level of cognitive load imposed by a particular task may fit better or worse with the

functionality offered by the search system. As a result, these two types of cognitive load (system-based and task-based) may either positively or negatively interact. Future studies will also help us better understand this important issue.

5. Conclusions

The gap between users' medical terminology and the terminology used in health information retrieval databases makes it difficult for laypersons to effectively find health information in these systems. Medical thesauri and MeSH tree browsers are helpful tools, but are not directly available for users to access in their information retrieval process. We addressed this issue by developing a system (MeshMed) with integrated MeSH Term Browser and MeSH Tree Browser components. In this study we compared MeshMed to a simple-interface system without these components (SimpleMed).

We proposed three research goals for this study. For the first goal, "to compare the effectiveness of MeshMed to SimpleMed in supporting consumers' health information retrieval," we found that explicitly integrating the two MeSH help components into the system was more effective in terms of perceived topic familiarity increase and question-answer performance. For the second goal, "to examine the roles of the two new MeSH help components implemented in MeshMed towards effective information retrieval," we found that the two MeSH help components specifically contributed to topic familiarity increases and encouraged participants to spend a significant amount of time interacting with them. Finally, for the third goal, "to explore whether and how the two new MeSH help components influence users' search behaviors and search strategies," we found that the two MeSH help components enabled users to develop additional search strategies in health information retrieval.

Most participants explicitly reported their preference of MeshMed as opposed to SimpleMed. The most useful component was the Term Browser which is particularly helpful in understanding and developing new concepts. The Tree Browser provided a unique hierarchical tree structure for an overview of related concepts and their relationships. Based on these findings and observations, we suggested that health information retrieval systems should support visible, immediately accessible thesaurus helps to facilitate users' information seeking tasks.

In conclusion, results from our study improved our knowledge of consumers' health information retrieval behaviors and how vocabulary help tools impact retrieval effectiveness for those users. Some of our results, such as our finding that searchers learn by using structured terminological support, could be expanded to other types of information retrieval, and deserve further studies in the future.

Acknowledgements

The authors thank all participants for their participation. We thank Natalie Hazen Burclaff for her valuable suggestions and great help in proofreading the paper. We also thank Dr. Razia Azen for her reviewing of the statistical treatments used in the study. Finally, helpful comments from the anonymous reviewers were greatly appreciated.

Appendix A. Biomedical terminology definition questions

1. What is COPD?
2. What is cerebral palsy?
3. What is diabetic gastroparesis?
4. What are beta blockers?
5. What is estrogen replacement therapy?
6. What is heavy alpha chain disease?

References

- Aronson, A. R., & Rindfleisch, T. C. (1997). Query expansion using the UMLS Metathesaurus. *AMIA annual symposium* (pp. 485–489).
- Bhagal, J., Macfarlane, A., & Smith, P. (2007). A review of ontology based query expansion. *Information Processing and Management*, 43(4), 866–886.
- Blair, D. C., & Maron, M. E. (1985). An evaluation of retrieval effectiveness for a full-text document-retrieval system. *Communications of the ACM*, 28(3), 289–299.
- Brewer, W. F., & Treysen, J. C. (1981). Role of schemata in memory for places. *Cognitive Psychology*, 13, 207–230.
- Cline, R. J. W., & Haynes, K. M. (2001). Consumer health information seeking on the Internet: The state of the art. *Health Education Research*, 16(6), 671–692.
- Coiera, E. W., & Vickland, V. (2008). Is relevance relevant? User relevance ratings may not predict the impact of internet search on decision outcomes. *Journal of the American Medical Informatics Association*, 15(4), 542–545.
- Efthimiadis, E. N. (1996). Query expansion. *Annual Review of Information Science and Technology*, 31, 121–187.
- Fidel, R., & Efthimiadis, E. N. (1995). Terminological knowledge structure for intermediary expert systems. *Information Processing and Management*, 31(1), 15–27.
- Fox, S. (2006). *Online health search 2006*. <http://www.pewinternet.org/~media/Files/Reports/2006/PIP_Online_Health_2006.pdf.pdf>. Retrieved 29.07.10.
- Gault, L. V., Schultz, M., & Davies, K. (2002). Variations in Medical Subject Headings (MeSH) mapping: From the natural language of patron terms to the controlled vocabulary of mapped lists. *Journal of Medical Library Association*, 90(2), 173–180.
- Gobel, G., Andreatta, S., Masser, J., & Pfeiffer, K. P. (2001). A MeSH based intelligent search intermediary for consumer health information systems. *International Journal of Medical Informatics*, 64(2–3), 241–251.
- Greenberg, J. (2001). Automatic query expansion via lexical-semantic relationships. *Journal of the American Society for Information Science*, 52(5), 402–415.

- Hersh, W. (2009). *Information Retrieval – A Health and Biomedical Perspective* (3rd ed.). New York: Springer.
- Hersh, W., Price, S., & Donohoe, L. (2000). Assessing thesaurus-based query expansion using the UMLS Metathesaurus. In *Proceedings of AMIA symposium*.
- Hersh, W. R., Hickam, D. H., Haynes, R. B., & McKibbin, K. A. (1994). A performance and failure analysis of SAPHIRE with a MEDLINE test collection. *Journal of the American Medical Informatics Association*, 1(1), 51–60.
- Herskovic, J. R., Tantaka, L. Y., Hersh, W., & Bernstam, E. V. (2007). A day in the life of PubMed: Analysis of a typical day's query log. *Journal of American Medical Informatics Association*, 14(2), 212–220.
- Kelly, D., & Cool, C. (2002). The effects of topic familiarity on information search behavior. In G. Marchionini, & W. Hersh (Eds.), *JCDL 2002. Proceedings of the second ACM/IEEE-CS joint conference on digital libraries* (pp. 74–75).
- Keselman, A., Browne, A. C., & Kaufman, D. R. (2008). Consumer health information seeking as hypothesis testing. *Journal of the American Medical Informatics Association*, 15, 451–484.
- Kim, K.-S., & Allen, B. (2002). Cognitive and task influences on web searching behavior. *Journal of the American Society for Information Science and Technology*, 53(2), 109–119.
- Kristensen, J. (1993). Expanding end-users' query statements for free text searching with a search-aid thesaurus. *Information Processing and Management*, 29(6), 733–744.
- Lu, Z., Kim, W., & Wilbur, W. J. (2009). Evaluation of query expansion using MeSH in PubMed. *Information Retrieval*, 12, 69–80.
- Marchionini, G., Dwiggins, S., Katz, A., & Lin, X. (1993). Information seeking in full-text end-user-oriented search systems: The roles of domain and search expertise. *Library & Information Science Research*, 15(1), 35–69.
- Marchionini, G., & Mu, X. (2003). User studies informing E-Table interfaces. *Information Processing and Management*, 39(4), 561–579.
- O'Brien, H. L., & Toms, E. G. (2008). What is user engagement? A conceptual framework for defining user engagement with technology. *Journal of the American Society for Information Science and Technology*, 59(6), 938–955.
- Ogilvie, P., & Callan, J. (2001). Experiments using the Lemur toolkit. In *Proceedings of the tenth text retrieval conference*.
- Plovnick, M. R., & Zeng, T. Q. (2004). Reformulation of consumer health queries with professional terminology: A pilot study. *Journal of Medical Internet Research*, 6(3), e27.
- Poikonen, T., & Vakkari, P. (2009). Lay persons' and professionals' nutrition-related vocabularies and their matching to a general and a specific thesaurus. *Journal of Information Science*, 35(2), 232–243.
- Raghavarao, D., & Padgett, L. V. (2005). *Block designs: Analysis, combinatorics, and applications*. New Jersey: World Scientific.
- Shiri, A. A., & Revie, C. (2003). The effects of topic complexity and familiarity on cognitive and physical moves in a thesaurus-enhanced search environment. *Journal of Information Science*, 29(6), 517–526.
- Shultz, M. (2006). Mapping of medical acronyms and initialisms to Medical Subject Headings (MeSH) across selected systems. *Journal of the Medical Library Association*, 94(4), 410–414.
- Soergel, D., Tse, T., & Slaughter, L. (2004). Helping healthcare consumers understand: An "interpretive layer" for finding and making sense of medical information. *Medinfo*, 931–935.
- Srinivasan, P. (1996a). Query expansion and MEDLINE. *Information Processing and Management*, 32(4), 431–443.
- Srinivasan, P. (1996b). Retrieval feedback in MEDLINE. *Journal of the American Medical Informatics Association*, 3(2), 157–167.
- Steinbrook, R. (2006). Searching for the right search-reaching the medical literature. *New England Journal of Medicine*, 354(1), 4–7.
- Stokes, N., Li, Y., Cavedon, L., & Zobel, J. (2009). Exploring criteria for successful query expansion in the genomic domain. *Information Retrieval*, 12, 17–50.
- Tang, M.-C. (2007). Browsing and searching in a faceted information space: A naturalistic study of PubMed users' interaction with a display tool. *Journal of the American Society for Information Science and Technology*, 58(13), 1998–2006.
- Vakkari, P. (2002). *Subject knowledge source of terms and terms selection in query expansion: An analytical study.. Proceedings of the 24th European colloquium on IR research*. Berlin: Springer (pp. 110–123). Berlin: Springer.
- Voorhees, M. E. (1994). Query expansion using lexical-semantic relations. In *Proceedings of the 17th annual international ACM SIGIR conference*.
- White, R. W., & Marchionini, G. (2007). Examining the effectiveness of real-time query expansion. *Information Processing and Management*, 43(3), 685–704.
- Wildemuth, B. M. (2004). The effects of domain knowledge on search tactic formulation. *Journal of the American Society for Information Science and Technology*, 55(3), 246–258.
- Yeganova, L., Comeau, D. C., Kim, W., & Wilbur, W. J. (2009). How to interpret PubMed queries and why it matters. *Journal of the American Society for Information Science and Technology*, 60(2), 264–274.
- Zeng, Q. T., Crowell, J., Plovnick, R. M., Kim, E., Ngo, L., & Dibble, E. (2006). Assisting consumer health information retrieval with query recommendations. *Journal of the American Medical Informatics Association*, 13(1), 80–90.
- Zeng, Q., Kogan, S., Ash, N., Greenes, R. A., & Boxwala, A. A. (2002). Characteristics of consumer terminology for health information retrieval. *Methods of Information in Medicine*, 41(4), 289–298.
- Zhang, J., Wolfram, D., Wang, P., Hong, Y., & Gillis, R. (2008). Visualization of health-subject analysis based on query term co-occurrences. *Journal of the American Society for Information Science and Technology*, 59(12), 1933–1947.